

Some commutative algebra

In order to proceed further, it is convenient to develop a bit of commutative algebra. In this section R will be a commutative ring (with identity).

Nilpotent elements

An element a of a ring R is said to be *nilpotent* if there is a positive integer n such that $a^n = 0$.

If $a^n = 0$, then we see that the following identity holds in the polynomial ring $R[T]$:

$$(1 - aT) \cdot (a^{n-1}T^{n-1} + \cdots + 1) = 1$$

In other words, $1 - aT$ is a unit in $R[T]$.

Conversely, suppose $(1 - aT) \cdot f(T) = 1$ in $R[T]$ for some $f(T) = \sum_{i=0}^k c_i T^i$. We compare coefficients of T^i to obtain:

$$\begin{aligned} c_0 &= 1 \\ c_i - ac_{i-1} &= 0 \text{ for } i = 1, \dots, k \\ -ac_k &= 0 \end{aligned}$$

which gives us $a^{k+1} = 0$. Thus, we have shown:

a is nilpotent in R if and only if $1 - aT$ is a unit in $R[T]$.

Hence, if a is not nilpotent in R , then $1 - aT$ is not a unit in $R[T]$. Hence, $1 - aT$ is contained in a maximal ideal M of $R[T]$. (One can prove this by induction if R is countable, otherwise one needs Zorn's lemma.) Note that the image of a in $R[T]$ is not in M . So a does not lie in the ideal

$$P = \{b \in R : \text{image of } b \text{ in } R[T] \text{ lies in } M\}$$

Now, R/P is a subring of the field $R[T]/M$ and is therefore domain. So P is a prime ideal in R .

Conversely, if there is a prime ideal P such that a does not lie in P , then a^n is not in P for all n . This means that which means a^n cannot be 0 for any n , so a is not nilpotent. Thus, we have shown:

a is not nilpotent in R if and only if there is a prime ideal P in R such that a does not lie in P .

Thus, the ideal-theoretic description of nilpotent elements is as follows:

The set (ideal) of nilpotent elements of a ring R is precisely the intersection of all prime ideals in R .

Primary ideals

An ideal I in a commutative ring is said to be P -primary for a prime ideal P , if:

- P contains I ;
- if Q is a prime ideal containing I then Q contains P .

Thus, P/I is the minimal prime ideal in the ring R/I and is therefore the intersection of all prime ideals in R/I . By what has been proved above P/I is precisely the collection of nilpotent elements of R/I . It follows that for every element a of P , there is a positive integer n so that a^n lies in I .

Given a P -primary ideal I and $a \in P$, there is an n such that a^n lies in I .

Intersection of primary ideals

Let $\{I_1, \dots, I_k\}$ be a finite set of primary ideals in a ring R and $I = \cap_{i=1}^k I_i$ be the intersection ideal.

Suppose that P_j is the prime ideal corresponding to the primary ideal I_j .

Given a prime ideal Q . If Q does not contain I_i , then it does not contain P_i , so P_i properly contains $Q \cap P_i$. Hence, there is an element a_i of P_i which is not in Q . As seen above, this means that there is an n_i such that $a_i^{n_i}$ lies in I_i . This means that the product $a_1^{n_1} \cdots a_k^{n_k}$ lies in I but does not lie in Q .

If a prime ideal Q does not contain each ideal in a finite collection of primary ideals, then Q does not contain their intersection.

Put differently:

If a prime ideal Q contains the intersection $\cap_{i=1}^k I_i$ of a finite collection of primary ideals, then Q must contain at least one of the I_i .

Maximal case. When $Q = M$ is a maximal, the image of a in R/M is invertible and thus there is an element b in R such that $1 - ab$ lies in M . In other words, we can write $1 = m + a'$ such that m lies in M and a' lies in I . By the binomial theorem, given any n , we see that we can write $1 = m' + a''$ where m' lies in M^n and a'' lies in I .

Given an element c of $M^n \cap I$, we write $c = c \cdot 1 = cm' + ca''$. Note that cm' and ca'' lie in $M^n \cdot I$. Thus, we see that $M^n \cap I = M^n \cdot I$.

When the prime ideals $P_i = M_i$ are all maximal we conclude:

For any collection of $\{M_1, \dots, M_k\}$ of maximal ideals and any positive integers $n_1, \dots, n_k$ we have $M_1^{n_1} \cap \cdots \cap M_k^{n_k} = M_1^{n_1} \cdots M_k^{n_k}$.

Primary ideals in Noetherian rings

When R is a finite ring, the set of elements of an ideal is finite. In particular, each ideal is *finitely generated* in R . A ring R for which every ideal is finitely generated is called a Noetherian ring.

So if I is a P -primary ideal in such a ring, then there is a *common* n such that a^n lies in I for *all* the generators of P . Using the binomial theorem, it follows that that $P^n \subset I$.

If I is a P -primary ideal in a Noetherian ring, then there is some positive integer n so that P^n is contained in I .

If a P -primary ideal I contains the intersection $\cap_{i=1}^k I_i$ of a finite collection of primary ideals, then P contains this intersection. As seen above at least one of the ideals I_i is contained in P . It follows that I_i^n is contained in I .

If a primary ideal I in a Noetherian ring contains the intersection $\cap_{i=1}^k I_i$ of a finite collection of primary ideals, then for at least one i , we have some power I_i^n of I_i which is contained in I .